The schedule reduction for the maximum number of stable matchings of order six

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Abstract

We reduce the determination of $f(6)$, the maximum number of stable matchings of a 6×6 stable-marriage instance, to a finite enumeration of *swap schedules*. The reduction rests on one combinatorial lemma (the bridge lemma) plus routine symmetry arguments; the enumeration itself (26.6 billion nodes) yields a maximum of 48, matching the dihedral lower bound, whence $f(6) = 48$.

1 Preliminaries

An *instance* I of order n consists of strict preference orders for n men over n women and vice versa. A perfect matching μ is *stable* if no pair (m, w) with $w \neq \mu(m)$ has both m preferring w to $\mu(m)$ and w preferring m to $\mu^{-1}(w)$. Let $\mathrm{sc}(I)$ be the number of stable matchings and $f(n) = \max_I \mathrm{sc}(I)$.

We use the standard structure theory of the stable-matching lattice and its rotations; see Gusfield–Irving [1], Chapters 2–3. We record the facts we need.

Fact 1. *Call (m, w) a* stable pair *of I if some stable matching contains it. For each man m, order his stable partners by his preference; this sequence is his* trajectory *T_m: the man-optimal matching μ_0 gives each man the first entry of his trajectory, and the rotations of I, applied from μ_0 in any linear extension of the rotation poset, move each man strictly down his trajectory, one entry per rotation containing him, ending at the woman-optimal matching. Symmetrically each woman moves strictly up her trajectory T_w (her stable partners in reverse preference order). A rotation ρ is a cyclic sequence $(m_0, w_0), \ldots, (m_{k-1}, w_{k-1})$, $k \geq 2$, of pairs of the current matching, whose application gives m_i the current partner w_{i+1} of m_{i+1} (indices mod k).*

Fact 2. *Every rotation moves $k \geq 2$ men. Each man is moved by exactly $|T_m| - 1 \leq n - 1$ rotations, so the total move count $\sum_\rho |\rho| = \sum_m (|T_m| - 1) \leq n(n-1)$; for $n = 6$: at most 30, and at most 5 moves per man.*

2 Schedules and read-off instances

Definition 1 (Schedule)**.** *Fix $n = 6$ and start from the identity matching $\mu_0(m) = m$. A* schedule *is a sequence of* steps*, each a cyclic move on $k \geq 2$ men (man m_i takes the current wife of m_{i+1}), subject to: total move count ≤ 30, at most 5 moves per man, and no man ever moving to a wife he has already had, nor any woman receiving a husband she has already had.*

The steps applied so far give each person a trajectory *(the sequence of partners, starting with the identity partner).*

Definition 2 (Read-off instance)**.** *Given a schedule S, the read-off instance $R(S)$ has: for each man, his trajectory in order as the top of his preference list, followed by the remaining women in a fixed canonical order; for each woman, her trajectory* reversed *as the top of her list, followed by the remaining men canonically.*

Lemma 1 (Validity)**.** *For any instance I of order 6, relabel women so that the man-optimal matching is the identity. Then the rotations of I, listed in any linear extension of the rotation poset, form a schedule $S(I)$ in the sense of Definition 1.*

Proof. Rotations are cyclic moves on $k \geq 2$ men exposed in the current matching (Fact 1); applying them in a linear extension of the poset from μ_0 realizes exactly the lattice traversal, in which men move strictly down and women strictly up their trajectories (hence no repeats on either side). The budget and per-man caps are Fact 2. $\square$

Lemma 2 (Bridge)**.** *For every instance I of order 6 (women relabelled as above),* $\mathrm{sc}(I) \leq \mathrm{sc}\big(R(S(I))\big)$*.*

Proof. Write $J = R(S(I))$. The trajectories of $S(I)$ are exactly the stable-partner sequences of I (Fact 1); thus for each man m, his J-list ranks his I-stable partners in the same relative order as his I-list (trajectory order is his I-preference order restricted to stable partners), with all other women strictly below; symmetrically for women.

We claim every I-stable matching μ is J-stable; since distinct matchings remain distinct, this proves the inequality. Note every pair of μ is a stable pair, hence a trajectory pair on both sides. Consider any (m, w) with $w \neq \mu(m)$.

Case 1: $w \notin T_m$. In J, m ranks w below all trajectory entries, in particular below $\mu(m) \in T_m$. So m does not prefer w; the pair does not block.

Case 2: $w \in T_m$. Then (m, w) is a stable pair of I, so also $m \in T_w$. Both w and $\mu(m)$ lie in T_m, and both m and $\mu^{-1}(w)$ lie in T_w (partners in stable matchings are stable pairs). Since J preserves the relative I-order within trajectories on both sides, (m, w) blocks μ in J if and only if it blocks μ in I — and it does not, as μ is I-stable.

Hence μ is J-stable. $\square$

Lemma 3 (Symmetry soundness)**.** *(i) Exchanging two adjacent steps whose man-sets are disjoint yields a schedule with the same trajectories, hence the same read-off instance. (ii) Relabelling men and women by a common permutation σ maps schedules to schedules and read-off instances to instances of equal stable count. Consequently, an enumeration that (a) requires the new men introduced by each step to be exactly the next unused labels (as a set), and (b) rejects a step that is lex-smaller than some earlier step from which it is separated only by steps man-disjoint from it, still attains the maximum of* $\mathrm{sc}(R(S))$ *over all schedules S.*

Proof. (i) Steps with disjoint man-sets touch disjoint women as well (their women are the current wives of their men), so each person's trajectory is unchanged by the exchange. (ii) Immediate. For the consequence: given any schedule, first apply (ii) with the greedy σ that assigns labels in order of first participation (any assignment within a step's new set, which rule (a) permits); the result satisfies (a) at every step. Rule (b) only discards a sequence in favour of a commutation-equivalent lex-smaller one, available by finitely many exchanges of type (i), which preserve $R(S)$. $\square$

Formalization status. Lemmas 1 and 2 are formalized in Lean 4 in the accompanying development (nine files from `SixBridge.lean` to `WRelabel6.lean`, ~4,600 lines; no `sorry`, axioms `propext/Classical.choice/Quot.sound`). The bridge is formalized in a bottom-agnostic form (`AbsBridge6.lean`): any instance that ranks each person's stable partners on top in original order dominates the original in stable-matching count, whatever its bottoms. The validity lemma is formalized as an explicit schedule: a cover chain from the man-optimal to the woman-optimal matching, built by repeatedly moving to the strict dominator of minimum rank-sum (`Chain6.lean`); each cover difference is decomposed into disjoint cyclic swaps by orbit extraction (`Cycle6.lean`); the concatenation is a legal schedule whose trajectories are exactly the stable-partner sequences (`ChainSched6.lean`); and a women-only relabelling normalizes the man-optimal matching to the identity (`WRelabel6.lean`). The composed statement `validity_unconditional` reads: for every well-formed instance I there is a legal schedule S with $\mathrm{sc}(I) \leq \mathrm{sc}(R(S))$. Lemma 3 concerns only the enumeration's internal symmetry reduction; its relabelling half is proved at the instance level (`Sym6.lean`) and the rest is validated computationally (canonicalization on/off agreement).

3 The theorem

Theorem 1. *Assume the exhaustive enumeration of all schedules of Definition 1 (with the reductions of Lemma 3, evaluating* $\mathrm{sc}(R(S))$ *at every node) attains maximum* 48*. Then* $f(6) = 48$*.*

Proof. Lower bound: the dihedral Latin instance has 48 stable matchings (direct computation; OEIS A351413). Upper bound: for any instance I, $\mathrm{sc}(I) \leq \mathrm{sc}(R(S(I)))$ by Lemmas 1 and 2; $S(I)$ is a node of the enumeration by Lemma 3, so $\mathrm{sc}(R(S(I))) \leq 48$. □

Remark 1. *The enumeration (`gen_enum.c`) traverses* 2.66×10^{10} *schedule nodes. An earlier pass evaluating only extension-maximal schedules (*1.40×10^{10} *exact stable-matching counts) also attained* 48*; the every-node pass removes any reliance on monotonicity of the read-off count under extension. Certification of the enumeration (independent reimplementation, and ultimately a verified replay in a proof assistant) is the remaining engineering step; the lemmas above are elementary given the standard rotation theory of [1].*

References

[1] D. Gusfield and R. W. Irving, *The Stable Marriage Problem: Structure and Algorithms*, MIT Press, 1989.